

Quantum computing with neutral atoms: assignment

Loïc Michalon, Hugo Mayer, Noé Mathieu

January 2025

1 Modeling the Antenna Frequency Problem as a Graph Coloring Problem on Unit-Disk Graphs

The antenna frequency assignment problem can be effectively modeled as a *graph coloring problem* on a *unit-disk graph*, as follows:

1.1 Nodes as Antennas

Each antenna in the system is represented by a node in the graph. The physical coordinates of the antennas determine the placement of the corresponding nodes in a two-dimensional plane.

1.2 Interference Criteria

Two antennas interfere with each other if the distance between them is less than a predefined threshold (8.7 km in this case). This is represented by an edge connecting the two corresponding nodes in the graph.

1.3 Graph Coloring

The problem is then transformed into a graph coloring task:

- Assign a color (frequency) to each node (antenna) in the graph.
- Ensure that two nodes connected by an edge (interfering antennas) are assigned different colors.

Minimizing the number of colors used corresponds to minimizing the number of frequencies required, which is the ultimate goal.

1.4 Unit-Disk Graph Representation

In this representation, each node in the graph is associated with a disk of radius 8.7 km centered at the node's coordinates. An edge is drawn between two nodes if their disks overlap, capturing the interference condition geometrically. This structure defines the graph as a unit-disk graph, which is a well-studied model for spatial constraints in wireless networks.

1.5 Conclusion

By reformulating the antenna frequency problem as a graph coloring problem on a unit-disk graph, we can leverage quantum algorithms to explore efficient solutions while adhering to the interference constraints and minimizing the number of frequencies used.

2 Designing an Analog Quantum Algorithm for Graph Coloring

To solve the graph coloring problem, we propose an analog quantum algorithm based on neutral atoms and adiabatic evolution. This algorithm leverages the identification of Maximum Independent Sets (MIS) as a building block for assigning colors to the graph's nodes. The approach is inspired by the techniques studied in TP3.

2.1 Main Idea

A *Maximum Independent Set (MIS)* is a subset of nodes in a graph such that:

- No two nodes in the subset are connected.
- No additional nodes can be added to the subset without violating the independence condition.

In the context of graph coloring:

- Each color corresponds to an independent set.
- Assigning a color to all nodes in a MIS ensures no conflicts (i.e., no edge between two nodes of the same color).
- The goal is to cover all nodes of the graph with the minimum number of colors.

2.2 Proposed Algorithm: Adiabatic Evolution for MIS Identification

2.2.1 Neutral Atom Modeling

Each node in the graph is represented by a neutral atom positioned according to its coordinates. Interactions are introduced between atoms corresponding to edges in the graph, encoding the problem's constraints.

2.2.2 Hamiltonians

The algorithm uses two key Hamiltonians:

- **Initial Hamiltonian** $H(0)$:

$$H(0) = |\delta| \sum_i n_i + \sum_{i,j} \frac{C}{r_{ij}^6} n_i n_j$$

This Hamiltonian represents a simple system where all nodes are deactivated ($n_i = 0$).

- **Final Hamiltonian** $H(T)$:

$$H(T) = -|\delta| \sum_i n_i + \sum_{i,j} \frac{C}{r_{ij}^6} n_i n_j$$

The final Hamiltonian favors configurations that correspond to a maximum independent set. The $-|\delta|n_i$ terms encourage node activation ($n_i = 1$), while the interaction term penalizes conflicts between connected nodes.

2.2.3 Adiabatic Evolution Process

1. The system starts in the ground state of $H(0)$, where all nodes are deactivated.
2. By slowly transforming $H(0)$ into $H(T)$ according to the adiabatic theorem, the system remains in the ground state throughout the evolution.
3. The final state corresponds to a maximum independent set in the graph.

2.2.4 Iterative Process for Graph Coloring

After identifying the first MIS:

1. Nodes in the MIS are assigned a color and removed from the graph.
2. The process is repeated on the reduced graph to identify additional MIS, with each corresponding to a new color.

3 Problem Modeling and Algorithm Justifications

3.1 Problem Modeling

The problem involves assigning frequencies to antennas in such a way that interference between antennas is minimized, and the total number of frequencies used is minimized. This problem can be translated into a graph coloring problem:

- **Nodes:** Each antenna is represented by a node in the graph.
- **Edges:** An edge is created between two nodes if their corresponding antennas are within the interference distance, meaning they cannot use the same frequency.
- **Goal:** The goal is to assign a color (frequency) to each node such that no two connected nodes share the same color, while minimizing the total number of colors used.

This graph representation allows the problem to be approached using well-established graph coloring techniques. In our case, we focus on an iterative solution based on Maximum Independent Sets (MIS), leveraging quantum adiabatic evolution to identify these sets efficiently.

3.2 Algorithm Choice

The algorithm is based on the identification of Maximum Independent Sets (MIS) through adiabatic evolution in a system of neutral atoms. This approach was chosen for the following reasons:

- **MIS as a Building Block:** An MIS corresponds to a subset of nodes that can all be assigned the same color. By identifying MIS iteratively, the entire graph can be colored.
- **Adiabatic Evolution:** This quantum analog approach exploits the adiabatic theorem to evolve the system from a simple Hamiltonian (representing a trivial configuration) to a complex Hamiltonian encoding the MIS constraints. The final state of the system corresponds to an MIS of the graph.
- **Neutral Atoms:** The use of neutral atoms allows the encoding of node interactions (edges) through controlled interactions between the atoms, making it well-suited for this graph-based problem.

The iterative process ensures that all nodes are colored, with each iteration producing a valid independent set that corresponds to a unique color.

3.3 Justifications of the Choices Made

The following aspects justify the choices made in the algorithm and its implementation:

- **Efficiency of MIS for Coloring:** Identifying independent sets is an efficient way to assign colors, as it ensures all nodes in the set can share the same color without conflict. Iteratively solving for MIS covers the graph systematically.
- **Adiabatic Theorem for Robustness:** The adiabatic theorem ensures that if the Hamiltonian is changed slowly enough, the system remains in its ground state. This makes the approach robust to noise and imperfections, as the system naturally converges to a valid solution.
- **Scalability with Neutral Atoms:** Neutral atom systems can scale to represent larger graphs, making the method applicable to real-world problems with a significant number of antennas (nodes).
- **Flexibility in Implementation:** The use of Pulser and QutipEmulator allows fine-tuning of pulse parameters and provides a flexible simulation environment to adapt the algorithm to various graph configurations.
- **Alignment with Physical Constraints:** By modeling nodes and interference distances using physical distances in the neutral atom system, the algorithm naturally aligns with real-world constraints, ensuring feasibility.

This approach balances theoretical rigor with practical implementation capabilities, demonstrating the potential of analog quantum computing for solving combinatorial optimization problems.

3.4 Limitations and Sub-Optimality

While this approach provides a valid solution to the graph coloring problem, it does not guarantee the most optimal result for the following reasons:

- **Greedy Nature of MIS Iteration:** The algorithm identifies MIS iteratively and assigns a unique color to each set. This greedy approach does not account for overlapping possibilities between MIS in successive iterations, which could reduce the total number of colors used.
- **Heuristic Dependence:** The algorithm relies on the adiabatic evolution process to approximate the MIS. This process can be sensitive to parameter choices (e.g., pulse duration, detuning), leading to suboptimal or locally optimal MIS solutions instead of the global optimum.
- **Adiabatic Constraints:** The speed of evolution is limited by the adiabatic theorem, which requires a slow transition to maintain the system in its ground state. Faster transitions might lead to errors, while slower transitions increase computation time, limiting practicality for large graphs.
- **No Explicit Optimization of Colors:** The algorithm minimizes conflicts by identifying MIS but does not explicitly optimize the coloring process to minimize the total number of colors. This results in a valid but potentially non-minimal coloring.

Despite these limitations, the approach showcases the potential of analog quantum algorithms for tackling combinatorial problems, providing valid solutions efficiently in a manner aligned with the physical constraints of the system.